

Institutional Vehicle Booking and Usage Register

A practical solution for transport management

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1. The Problem & Who is Affected

Problem:

- Vehicles are booked via phone calls and a paper diary.
- Double bookings are common and discovered too late.
- No record of actual usage, making it impossible to attribute costs to departments.

Who is affected:

- The Transport Clerk (stressful double-bookings).
- Departments (delayed or missing vehicles).
- The Transport Officer (no data for cost tracking).

2. Our Solution

A digital register that securely records bookings, automatically blocks overlapping requests, and instantly calculates distance usage per department.

3. Capture Booking

Institutional Vehicle Booking & Usage Register

Transport Office — PDKVCET

online

Book a Vehicle

Register

Usage by Department

VEHICLE

Select vehicle



DEPARTMENT

e.g. AIDS

REQUESTED BY

Full name

JOURNEY DATE

07/27/2026



START TIME

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END TIME

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PURPOSE

e.g. Guest pickup - airport

Save booking

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4. Register View

Institutional Vehicle Booking & Usage Register

Transport Office — PDKVCET

online

Book a Vehicle

Register

Usage by Department

Search vehicle, name, department, purpose...

All statuses



ID	VEHICLE	REQUESTED BY	DEPT	DATE	TIME	PURPOSE	KM	STATUS
BK-0041	TN22AC5678	GEORGE	AIDS	2026-07-28	08:07-21:09	OFFICE WORK	not recorded	Pending
BK-0015	TN22AE3456	M. Suresh	Sports	2026-07-26	11:00-14:00	Faculty conference travel	not recorded	Completed
BK-0010	TN22AD9012	T. Bala	Administration	2026-07-24	10:00-13:30	Faculty conference travel	not recorded	Pending
BK-0030	TN22AC5678	R. Karthik	Civil	2026-07-24	09:00-10:00	Faculty conference travel	not recorded	Pending
BK-0008	TN22AC5678	K. Meena	Administration	2026-07-20	11:00-14:00	Material collection - vendor	26.3	Completed
BK-0002	TN22AB1234	R. Karthik	AIDS	2026-07-20	10:00-13:00	Exam paper collection	88.2	Completed
BK-0025	TN22AD9012	S. Priya	Mechanical	2026-07-20	10:00-13:30	Material collection - vendor	not recorded	Pending
BK-0040	TN22AD9012	N/A	Unassigned	2026-07-20	10:00-13:00	misc	not recorded	Pending

5. Usage Dashboard & Derived Figures

Derived figures (Total KM and Trip Count) are calculated on the server. Bookings without KM yet are excluded from total distance but included in trip count.



6. What Works & Unfinished Features

What Works:

- Full end-to-end booking capture and validation.
- Server-side double booking prevention.
- Real-time usage aggregation by department.
- Offline mode (queues bookings and syncs when reconnected).

Unfinished:

- No authentication/login system.
- No edit or cancel UI directly from the frontend.

7. Next Improvement

Next Step: Add a vehicles table (make, capacity, fuel type) to calculate the actual financial cost per department, not just the distance traveled.